Topic 4: Computers

Students must be familiar with the hardware and software components that make up a computer system and recognise that computers come in all shapes and sizes from embedded microprocessors to distributed clouds.

Students should be given the opportunity to gain practical experience of interpreting instructions written in assembly language.

Subject content	Students should:
4.1 Hardware	4.1.1 Understand the function of hardware components of a computer system [processor (CPU), memory, secondary storage, input, output] and explain how they work together
	4.1.2 Understand the concept of a stored program and the role of components of the processor [control unit (CU) arithmetic/logic unit (ALU), registers, clock, address bus, data bus] in the fetch-decode-execute cycle
	4.1.3 Be able to interpret a block of assembly code using a given set of commands *
	4.1.4 Understand that there is a range of different input and output devices available and be able to select appropriate input/output devices for a specified purpose
	4.1.5 Understand how data is stored on physical devices [magnetic,